



Paper ID : 261279

Printed Page: 1 of 1

Subject Code: MPC103T

Roll No:

MPHARM**(SEM. I) THEORY EXAMINATION 2025-26****ADVANCED MEDICINAL CHEMISTRY****TIME: 3 HRS****M.MARKS: 75****Note:** 1. Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2x10 = 20**

(a)	What is the full form and therapeutic significance of COX-2 inhibitors?
(b)	Define Pharmacophore identification in lead optimization.
(c)	How does Optical Isomerism affect the metabolism of a drug?
(d)	What is the rationale behind designing a Prodrug?
(e)	Name one novel target for anticancer therapy other than traditional DNA alkylators.
(f)	Define Artificial Enzymes.
(g)	Mention the mechanism of action of Vortioxetine.
(h)	What are the common causes of Antibiotic Resistance?
(i)	Define the term Target Validation.
(j)	Name one Anti-HIV agent from the Integrase Inhibitor class.

SECTION B**2. Attempt any two of the following:****2 x 10 = 20**

(a)	Discuss the impact of Stereoisomerism (Optical and Geometrical) on the pharmacokinetic and pharmacodynamic profiles of drugs with suitable examples.
(b)	Explain the Basic Concepts of Prodrug Design. Discuss how prodrugs are used to improve drug absorption and reduce toxicity.
(c)	Provide a systematic study of Oral Hypoglycemic Agents. Discuss the SAR and mechanism of action of DPP-IV inhibitors like Omargliptin.

SECTION C**3. Attempt any five part of the following:****7 x 5 = 35**

(a)	Discuss the role of STAT-3 (Signal Transducer and Activator of Transcription-3) as a novel molecular target in drug discovery.
(b)	Explain Analog Design using Bioisosteric replacement and Rigid Analogs as strategies.
(c)	Explain the design and therapeutic applications of Peptidomimetics.
(d)	Explain COX-1 and COX-2 inhibitors with therapeutic significance.
(e)	Discuss the Rational Design of Covalently Binding Enzyme Inhibitors with examples.
(f)	Write a note on the systematic study and synthesis of new generation H ₂ Receptor Antagonists like Lafutidine.
(g)	Describe the SAR and mechanism of action of New Generation Adrenergic Agents (e.g., Celiprolol).